Analysis of Toxicity in Online Political Discussions

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***Abstract*—This study compares a baseline TF-IDF logistic regression model with a pretrained transformer model (Toxic BERT) for toxicity classification, and examines how differences in model predictions manifest in real-world political discourse on Reddit. Logistic regression achieved strong performance (ROC AUC = 0.965, F1 = 0.767), but Toxic-BERT outperformed it across all metrics (ROC AUC = 0.998, F1 = 0.922). Despite these differences, both models identified nearly identical proportions of toxic content in Reddit posts (4.89% for logistic regression and 4.96% for Toxic-BERT). However, the models disagreed on 4.12% of posts, revealing systematic differences in how toxicity is interpreted. Logistic regression tends to rely on keyword-level signals, while Toxic-BERT incorporates context, particularly in distinguishing direct insults from reported or implied statements. Additional analyses show that these differences are not explained by observable covariates such as subreddit, post length, or engagement. These findings suggest that model disagreement arises from differences in language interpretation rather than data characteristics, with important implications for measuring and moderating toxicity in political communication.**

***Index Terms*—Toxicity, natural language processing, classification, logistic regression, transformer models, political communication, social media**

I. Introduction

For decades, natural language processing (NLP) methods have been widely used to analyze sentiment across a variety of contexts [1], [2]. More recently, research has emerged with a focus on toxicity, defined as language that is disrespectful or likely to discourage participation in discussion [3]. The measurement of toxicity in online political discourse has become particularly important for both research and content moderation, yet estimates of toxicity may depend strongly on the choice of modeling approach. This raises important concerns about the comparability of toxicity measurements. If different models systematically interpret the same text in different ways, then conclusions about the prevalence and distribution of toxicity may be sensitive to modeling assumptions rather than underlying behavior.

To measure toxicity, a range of modeling approaches have been developed, spanning from traditional methods to modern deep learning models [2]. Logistic regression remains a popular baseline model for text classification due to its simplicity and interpretability. In NLP applications, text is first transformed into numerical features, most commonly using Term Frequency–Inverse Document Frequency (TF-IDF), which weights words by their importance within and across documents. Prior studies have shown that logistic regression combined with TF-IDF can achieve strong performance in sentiment and toxicity classification, often serving as a reliable benchmark against more complex models [4].

In contrast, transformer attention-based models, such as BERT, first appeared in 2017 and represent state-of-the-art approaches, leveraging attention mechanisms and bidirectional context to capture complex linguistic relationships [5], [6]. It is important to understand the entire context of a statement, rather than focusing on key word methods as toxicity can appear in many forms, including direct insults, reported statements, and implied or indirect forms of demeaning language. As a result of these different model constructions, logistic regression and BERT may systematically differ in how they interpret and classify toxic language, particularly in cases where meaning depends on context rather than the presence of explicit keywords.

These differences are particularly important in the context of online political discourse [7], [8]. Exposure to toxic content can exacerbate partisan divisions [9]. Platforms such as Reddit, which organize discussion into communities based on shared interests and beliefs, may further amplify these dynamics. Additionally, prior research has found that political content containing inflammatory or hateful language is more likely to spread and receive engagement, creating incentives for provocative or toxic rhetoric [10]. Together, these findings suggest that both the structure of online platforms and the nature of political discourse contribute to the prevalence of toxicity.

The goal of this study is to compare a baseline logistic regression model and a transformer-based model (Toxic-BERT) and to understand how differences in their predictions manifest in real-world political discourse. By analyzing both predictive behavior and patterns of disagreement between models, this study aims to better understand how modeling choices influence the measurement and interpretation of toxicity. These findings have important implications for both social science research and the use of automated systems in content moderation.

II. Methods

The question is a supervised binary classification problem, where the input consists of text data (Reddit post titles or comments), and the output is a binary label indicating whether the content is toxic. The analysis proceeds in three stages: (1) training and evaluating models on labeled Jigsaw data, (2)

tuning classification thresholds, and (3) applying both models to Reddit data to analyze prediction patterns and disagreement.

A. Data Sources

The analysis combines two datasets. The primary dataset consists of 7,503 Reddit posts from political subreddits (2014-2020), including post titles and associated metadata such as subreddit, score, and number of comments [11]. Titles of posts were used instead of comments because they can be treated as approximately independent observations. In contrast, comments within a thread are not independent, as they are influenced by prior messages and user interactions. Modeling titles does not require accounting for network or conversational dependencies, which simplifies the modeling framework. The secondary dataset is the Jigsaw Toxic Comment Classification dataset (approximately 450,000 labeled observations) [12], which provides binary toxicity labels and is used to train the baseline logistic regression model and compare performance of this baseline and the Toxic BERT model.

B. Baseline Model: Regularized Logistic Regression

The data was preprocessed via tokenization and stopword removal. In order to use logistic regression, the text data is transformed into numerical features using term frequency-inverse document frequency (TF-IDF). The TF-IDF weight for term t in document d is defined as

$$\text{tfidf}(t, d) = \text{tf}(t, d) \log\left(\frac{N}{df(t)}\right), \tag{1}$$

where $\text{tf}(t, d)$ denotes term frequency, $df(t)$ denotes document frequency, and N is the total number of documents. The feature space is restricted to a fixed vocabulary (e.g., the top 10,000 terms) to reduce dimensionality.

Let $x_i \in \mathbb{R}^p$ denote the TF-IDF feature vector for observation i, and let $y_i \in \{0, 1\}$ denote the binary toxicity label. The logistic regression model specifies the conditional probability of toxicity as

$$P(y_i = 1 \mid x_i) = \frac{1}{1 + \exp\left(-\beta_0 - x_i^\top \beta\right)}, \tag{2}$$

where β_0 is the intercept and $\beta \in \mathbb{R}^p$ is the coefficient vector.

Model parameters are estimated via penalized maximum likelihood with L_2 regularization. The objective function is

$$\ell(\beta) = \sum_{i=1}^{n} \left[y_i \log(p_i) + (1 - y_i) \log(1 - p_i)\right] - \lambda \|\beta\|_2^2, \tag{3}$$

where $p_i = P(y_i = 1 \mid x_i)$ as defined in (2), and $\lambda > 0$ is a regularization parameter. To address class imbalance, we incorporated class weights into the likelihood function, assigning higher weight to the minority class. Specifically, each observation was weighted inversely proportional to its class frequency, so that misclassification of minority class observations incurred a larger penalty during estimation. This modifies the objective function to a weighted penalized likelihood, improving sensitivity to the toxic class. Model performance was evaluated using 5-fold stratified cross-validation with area under the ROC curve (AUC) as the scoring metric. After cross-validation, the model was fit on the full training set.

C. Transformer Model: Toxic BERT

To capture contextual dependencies between words, a pre-trained transformer model (Toxic BERT) is applied [13]. Let $h_{[\text{CLS}],i}$ denote the contextual embedding corresponding to observation i. The predicted toxicity probability is given by

$$\hat{p}_i = \text{softmax}\left(h_{[\text{CLS}],i} W + b\right), \tag{4}$$

where W and b are parameters of the classification layer. Input text is tokenized and truncated to a fixed maximum sequence length of 512 tokens. Predictions are generated in batches to improve computational efficiency and manage memory constraints as it enables parallel evaluation of multiple inputs when processing large datasets one observation at a time.

D. Model Evaluation and Comparison

Predicted probabilities were converted into binary classifications using model-specific thresholds. For the logistic regression model, the classification threshold was selected to maximize the F1-score using cross-validated predictions on the training data. For the Toxic-BERT model, threshold selection was performed on a separate validation subset of 10,000 labeled observations, distinct from the held-out evaluation set.

To enable direct comparison, both models were evaluated on the same held-out subset of 10,000 labeled Jigsaw observations. Performance was assessed using multiple metrics, including ROC AUC, precision, recall, and F1-score, to capture both overall ranking performance and the tradeoff between false positives and false negatives under class imbalance. This ensures that differences in performance reflect model behavior rather than differences in evaluation data or metrics. However, because this evaluation set was used in pretraining Toxic-BERT, its performance estimates are likely optimistic. Therefore, results should be interpreted as a comparison between a baseline model and a pretrained classifier, rather than a strictly unbiased out-of-sample evaluation for the transformer model.

E. Reddit Data Analysis

To compare model toxicity predictions in a real-world context, both models were applied to the Reddit dataset of political posts. Model disagreement was analyzed in two ways. First, binary disagreement was measured as the proportion of observations for which the two models produced different class predictions. Second, the magnitude of disagreement was assessed by examining the absolute difference in predicted probabilities between models.

To further investigate how toxicity predictions relate to post characteristics, additional analyses were conducted linking model outputs to covariates, including text length, post score, and number of comments. These analyses examine whether disagreement between models is systematically associated

with features of the content, providing insight into the contexts in which simpler linear models and transformer-based models diverge in their assessments of toxicity.

III. Results

A. Model Performance on Labeled Data

The logistic regression model demonstrated strong performance on the labeled Jigsaw dataset. The model achieved an average ROC AUC of 0.964, indicating high discriminative ability between toxic and non-toxic comments. The TF-IDF representation included 6,427 unigrams, 2,359 bigrams, and 1,214 trigrams. This suggests that while single-word features dominate, higher-order n-grams contribute meaningful additional information.

An optimal classification threshold of 0.772 was selected for the logistic regression model. This relatively high threshold indicates that the model requires strong evidence before classifying a comment as toxic, reflecting a tradeoff that prioritizes precision over recall. For the Toxic-BERT model, an optimal threshold of 0.495 was identified. This threshold is close to the default value of 0.5, suggesting that the pretrained model is well-calibrated and does not require substantial adjustment to balance precision and recall.

The distribution of predicted probabilities on the evaluation dataset further highlights differences in model behavior. Toxic-BERT assigns extremely low probabilities to the majority of observations (median = 0.00076), with a small subset of predictions near one, indicating highly confident classifications. In contrast, logistic regression produces a broader distribution of probabilities (median = 0.081), with substantially more mass in the intermediate range, reflecting greater uncertainty in its predictions.

Table I summarizes model performance on the held-out evaluation dataset. Toxic-BERT outperforms logistic regression across all metrics, achieving higher precision, recall, F1-score, and AUC. The improvement in precision (0.919 vs. 0.813) indicates that Toxic-BERT produces fewer false positives, while the higher recall (0.925 vs. 0.725) suggests it is also more effective at identifying toxic content. The gap between precision and recall for logistic regression suggests that the model is more prone to false negatives, whereas Toxic-BERT achieves a more balanced tradeoff between precision and recall. Both models achieve high AUC values, though the near-perfect AUC of Toxic-BERT reflects its strong ranking performance and potentially optimistic performance metrics due to using in-sample data for evaluation. These differences in performance and threshold behavior suggest that the models may exhibit distinct prediction patterns, which are further explored in the analysis of Reddit data.

TABLE I
Model performance on labeled Jigsaw evaluation data

Model	ROC AUC	Precision	Recall	F1
Logistic Regression	0.965	0.813	0.725	0.767
Toxic-BERT	0.998	0.919	0.925	0.922

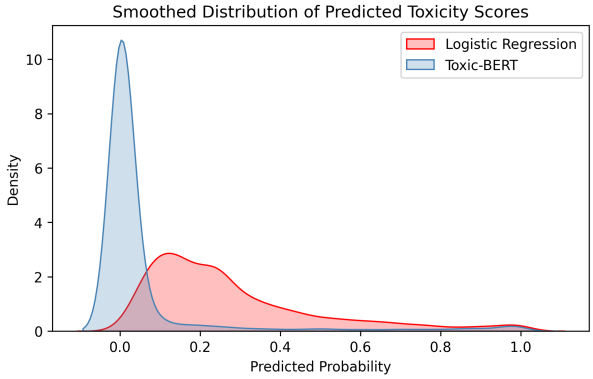

Fig. 1. Smoothed distributions of predicted toxicity probabilities for logistic regression and Toxic-BERT.

B. Toxicity Distributions and Model Disagreement on Reddit Data

At the chosen thresholds, Toxic-BERT classified approximately 4.96% of Reddit posts as toxic (372 total), while logistic regression classified 4.89% as toxic (367 total). Although the overall proportion of posts labeled toxic is nearly identical, the distribution of predicted toxicity scores in Figure 1 reveals important differences between the models. Toxic-BERT produces a distribution that is highly concentrated near zero, with a sharp peak at very low probabilities and relatively few high-confidence toxic predictions. In contrast, logistic regression yields a much more dispersed distribution, with substantial mass in the intermediate probability range (roughly 0.1-0.4) and a longer right tail. This suggests that logistic regression assigns many posts moderate toxicity probabilities, whereas BERT more often separates posts into very low-probability and higher-probability cases. This pattern is consistent with the evaluation dataset, where logistic regression appeared more uncertain overall and BERT exhibited stronger separation between toxic and non-toxic examples.

Prior to threshold tuning, logistic regression classified a much larger share of posts as toxic: 14.75% at the default 0.5 threshold, compared with 4.89% at the F1-optimal threshold of 0.77. After threshold adjustment, the proportion of posts labeled toxic by logistic regression dropped substantially and aligned much more closely with BERT. This highlights the importance of threshold selection, since the default cutoff can produce misleading comparisons when models have very different score distributions.

While the density plot highlights differences in overall score distributions, Figure 2 illustrates how these differences manifest at the level of individual posts. The threshold lines reveal regions where the models produce conflicting classifications, showing that disagreement arises for a nontrivial subset of observations. The two models disagreed on the toxicity label in 4.12% of cases (309 posts). Disagreement occurred in both directions, with 2.09% (157 posts) classified as toxic by Toxic-BERT but not by logistic regression, and 2.03% (152 posts) classified as toxic by logistic regression but not

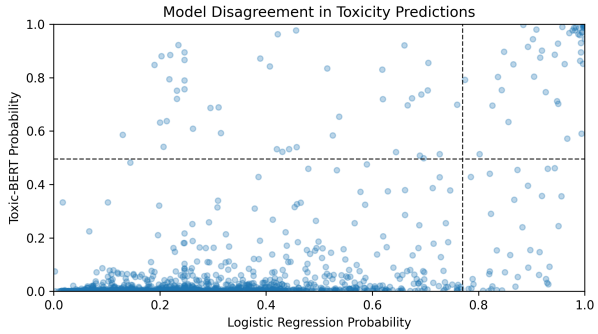

Fig. 2. Scatterplot of predicted toxicity probabilities from logistic regression and Toxic-BERT for a sample of 2,000 Reddit posts. The dashed vertical and horizontal lines indicate the classification thresholds for logistic regression (0.77) and Toxic-BERT (0.495).

by Toxic-BERT. This indicates that neither model consistently produces more toxic classifications than the other. Differences between models are not purely distributional and represent fundamentally different assessments of the same text.

Several illustrative examples highlight these differences, particularly in how the models handle profanity, keywords, and contextual tone. For example, a post referencing "execution" and "torturing prisoners" receives a very high toxicity score from logistic regression (0.92) but a near-zero score from BERT (0.03), suggesting that the logistic model reacts strongly to violent keywords even in a non-abusive historical frame. By contrast, a post describing a public figure as a "moocher" and "deadbeat" is assigned a relatively low probability by logistic regression (0.19) but a substantially higher probability by BERT (0.63), indicating that BERT may better capture demeaning tone even in the absence of canonical toxic keywords. Similarly, strongly accusatory language, such as labeling someone a "terrorist," tends to receive higher scores from BERT than from logistic regression, reflecting greater sensitivity to context and implied insult rather than vocabulary alone.

At the same time, there are cases in which BERT assigns relatively low toxicity scores to statements that might initially appear strongly negative. For example, the statement "Howard Dean calls Steve Bannon 'a Nazi'" receives a high score from logistic regression (0.78) but a much lower score from BERT (0.25). However, further inspection suggests that this behavior is not necessarily inconsistent. BERT appears to distinguish between direct assertions and reported speech: statements such as "X calls Y a Nazi" are treated as descriptive or attributed claims, whereas direct statements like "Y is a Nazi" or generalized attacks such as "all conservatives are Nazis" receive much higher toxicity scores.

All of these examples suggest that BERT is sensitive to whether language constitutes a direct insult or a reported characterization. In contrast, logistic regression assigns high toxicity scores in these cases due to the presence of strongly negative terms, regardless of how they are used. Overall, these patterns reinforce the interpretation that logistic regression behaves more like a keyword-based classifier, whereas BERT applies a more context-dependent interpretation of toxicity, even within politically charged language.

Cases in which both models classify posts as toxic are typically characterized by explicit profanity, direct insults, or clearly abusive language directed at individuals or groups. In these examples, both logistic regression and Toxic-BERT assign consistently high toxicity probabilities, often exceeding 0.8 or 0.9, indicating strong agreement in identifying overtly hostile content. This suggests that highly explicit or unambiguous toxic language is reliably detected by both models, and that their differences primarily emerge in more subtle or context-dependent cases rather than in clearly abusive statements.

C. Reddit Data Covariate Analysis

Figure 3 shows the proportion of posts classified as toxic by logistic regression and Toxic-BERT across subreddits. Both models identify r/The_Donald as having the highest share of toxic posts and r/SandersForPresident as among the lowest. Although BERT tends to assign slightly higher toxicity rates in several subreddits (e.g., r/The_Donald, r/Libertarian, and r/democrats), these differences are generally modest, typically on the order of one to two percentage points. In a few cases, such as r/politics and r/Conservative, logistic regression assigns slightly higher toxicity rates, but again the differences are small.

Taken together, these results indicate that both models agree on the overall distribution of toxicity across subreddits, suggesting that variation in toxicity is not primarily driven by model choice. This consistency across models also suggests that the types of toxic language present in these subreddits do not differ substantially. In particular, neither model consistently identifies markedly more toxicity than the other within any single community, indicating that there is no strong evidence that certain subreddits contain systematically different forms of toxicity that favor one modeling approach over the other.

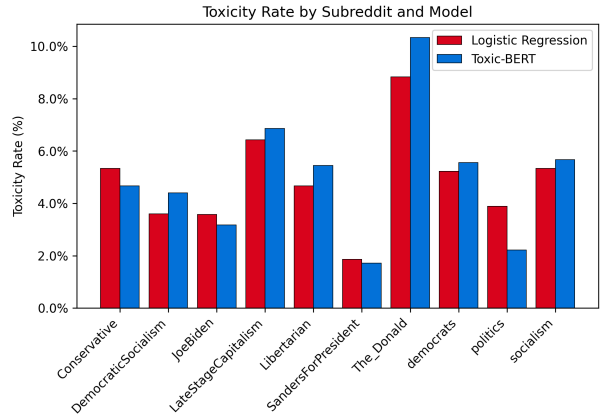

Fig. 3. Logistic Regression and Toxic-BERT Toxicity Classifications by Subreddit.

Toxicity rates and model disagreement were also examined across post length and engagement (measured by score). Across both covariates, patterns are generally stable and do not exhibit strong systematic variation. Toxicity rates for logistic regression and Toxic-BERT remain similar across length bins, with only modest differences and no clear trend indicating that longer or shorter posts consistently lead to greater divergence between models. Similarly, when posts are grouped by score, toxicity rates fluctuate slightly but do not display a consistent relationship with engagement. In both cases, the two models closely track one another, and any differences are small and irregular rather than systematic. Overall, there is little evidence that observable covariates meaningfully explain differences between the models. Instead, model disagreement appears to arise primarily from how each model interprets individual pieces of text rather than from systematic variation across subsets of the data.

IV. Conclusion

This study compared a baseline TF-IDF logistic regression model with a pretrained transformer-based model (Toxic-BERT) performance on toxicity classification, and examined how differences in model predictions manifest in political discourse on Reddit. While Toxic-BERT outperformed logistic regression on labeled evaluation data across all metrics, both models produced nearly identical overall toxicity rates when applied to real-world Reddit posts. Despite this aggregate similarity, the models disagreed on a nontrivial subset of observations, indicating systematic differences in how toxicity is identified.

Analysis of disagreement patterns suggests that logistic regression functions largely as a keyword-driven classifier, assigning high toxicity scores based on the presence of strongly negative or violent terms regardless of context. In contrast, Toxic-BERT demonstrates greater sensitivity to linguistic structure and framing, distinguishing between direct insults, reported speech, and more subtle forms of demeaning language. These differences are most apparent in edge cases, where contextual interpretation plays a critical role in determining whether content is considered toxic. Importantly, covariate analyses show that these differences are not explained by observable features such as subreddit, post length, or engagement. These findings suggest that model disagreement arises primarily from differences in how language is interpreted at the individual post level, rather than from systematic variation across subsets of the data.

The primary contribution of this study is to show that models with similar aggregate performance can differ substantially in how they interpret individual instances of toxicity. These findings also have broader implications for the study and moderation of political communication. Because different models can reach different conclusions about the same content, estimates of toxicity in political discourse may depend heavily on the modeling approach used. In particular, keyword-based models may overestimate toxicity in politically charged but non-abusive contexts, while context-sensitive models may under-detect harmful rhetoric that relies on implication or shared knowledge. This has important consequences for content moderation, measurement of polarization, and research on online incivility, as model choice can shape both what is identified as toxic and how prevalent toxicity appears to be.

Several limitations should be noted. The evaluation of Toxic-BERT may be optimistic due to overlap with its training data, and the Reddit dataset consists only of post titles, which may limit the ability to fully capture context. Future work could explore fine-tuning transformer models on social media and political interaction data or incorporating full comment threads. Overall, these results highlight the importance of contextual modeling in toxicity detection and demonstrate that differences between models are driven more by interpretation than by data characteristics.

Code Availability

The full implementation for this project is available at: https://github.com/enoracat/Stats_507_Final_Project

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